Does a paraglider move differently in a group?

Group time against solo flight, on open flights

3 November 2026

Three questions

1 Do flights in a group move differently from solo flights?

Raw

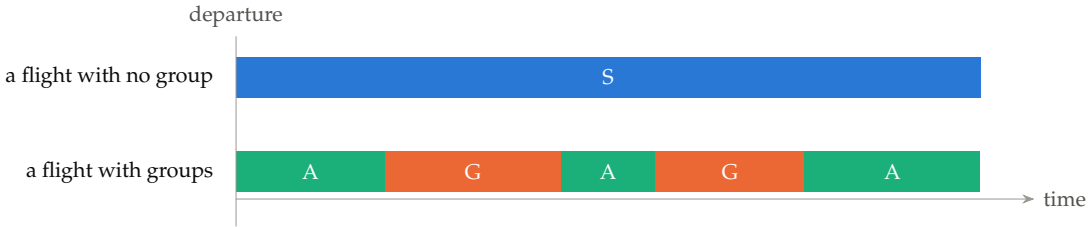
2 Is it the group, or the route, season, hour and wing that go with it?

Same route and factors

3 Does the same flight change when it enters a group, and when it leaves?

Same flight

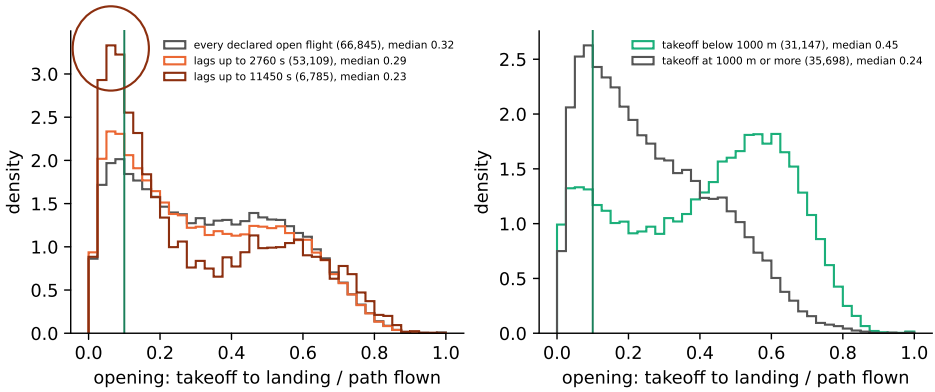
Three populations of flight time



- S (Solo flight): flights that never entered a group.
- G (Group flight): group time of the flights that entered one.
- A (Solo stretch): the solo part of the group flights.

A *piece* is an uninterrupted stretch of one population.

Which flights: the open ones

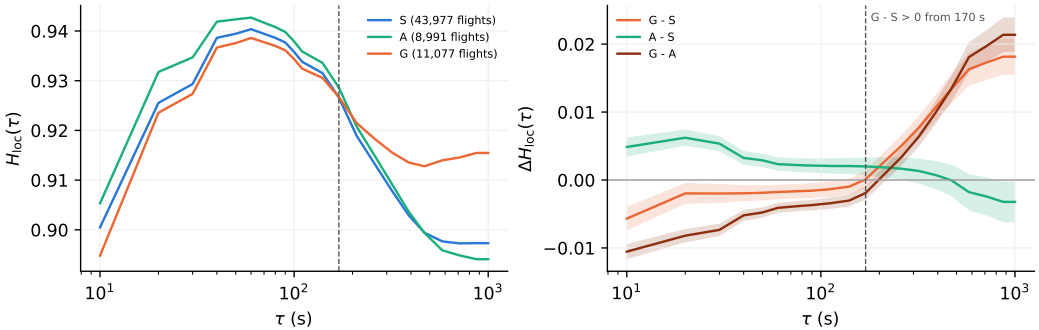


- **Opening** $= \dfrac{\|\mathbf{r}_{\text{land}} - \mathbf{r}_{\text{take}}\|}{\sum_i \|\mathbf{r}_{i+1} - \mathbf{r}_i\|}$: 1 straight, ≈ 0 back at the takeoff.
- We keep opening ≥ 0.1: 84% of the flights; 88% of low takeoffs, 80% of high ones (right).

Question 1

Do flights in a group move differently from solo flights?

Raw: group time is straighter at the long lags

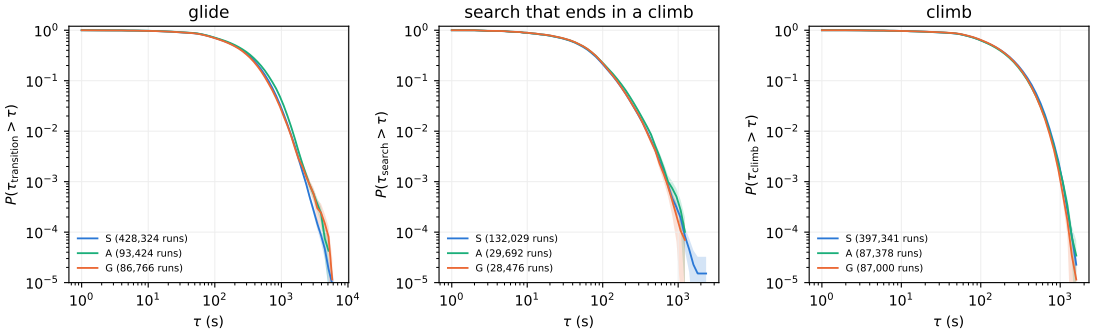


- $H_{\mathrm{loc}}(1000\ \mathrm{s})$: G 0.915, S 0.897, A 0.894.
- $\mathrm{G} - \mathrm{S} = +0.018\ [+0.015, +0.021]$; -0.006 at 10 s, positive from 170 s

At first the group waits to form, a solo pilot leaves at once. After one glide (median ≈ 200 s) group time gains, probably because group flights favour zones and days with more thermals.

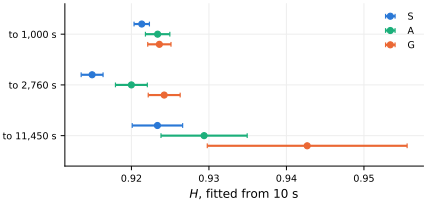
→ how we compute H_{loc} → the bands → longer lags → map of flights

Raw: the durations of the phases



- Climbs, searches and glides last as long in S, A and G: the survival curves overlap.

Fitted H of S, A and G

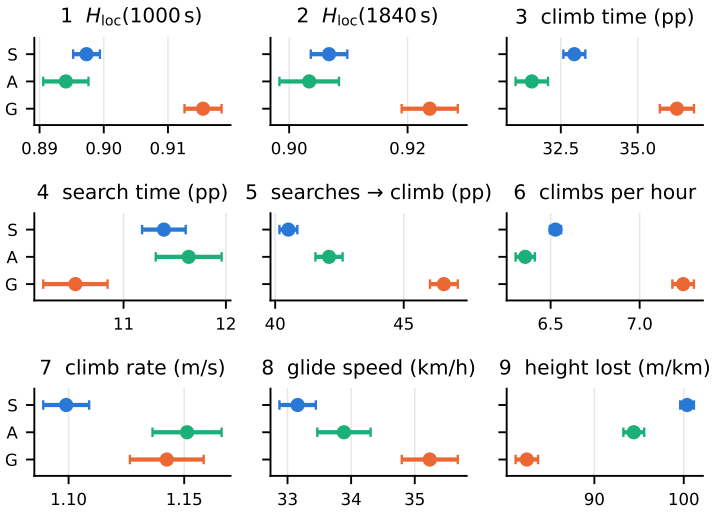


- H: slope of log MSD from 10 s to the end of the lags T; a mean of H_{loc} over the lags (within 0.002).
- Careful: H_{loc} is not constant (it varies by 0.04–0.05 along a curve, up to 0.09 at the longest lags): H is not a well-defined exponent and depends on T.

H of G exceeds that of S by 0.003 to 1000 s, 0.009 to 2760 s and 0.020 to 11 450 s (point values): the gap grows with the lags used.

→ lags to 2760 s
→ the longest lags
→ H_{loc} at the edge

Raw: group time also flies differently



3, 6 G climbs more and finds more climbs.

4, 5 G searches less and better.

7 A ≈ G: zones with stronger thermals, better conditions.

8 G keeps a higher glide speed (Williams et al. 2018).

9 G glides flatter.

pp: percentage points; 90% bands (5–95%). Bands of G − S exclude 0 in panels 3–9.

→ how each flight counts
→ the seven statistics

Question 2

Is it the group, or what goes with it?

The raw gap mixes everything

Group time differs from solo time in

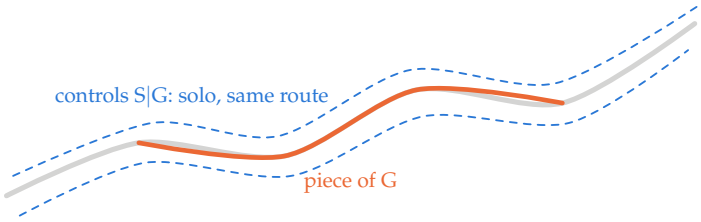
- the **route** flown,
- the **season** and the **hour**,
- the **wing**,
- the pilots who fly in groups.

We take them apart

1. fix them, and see what is left;
2. split the gap among them.

Same route, season, hour and wing: does the group still make a difference?

Same route, season, hour and wing



route: 80% of the piece within 500 m

another date, ±2 months

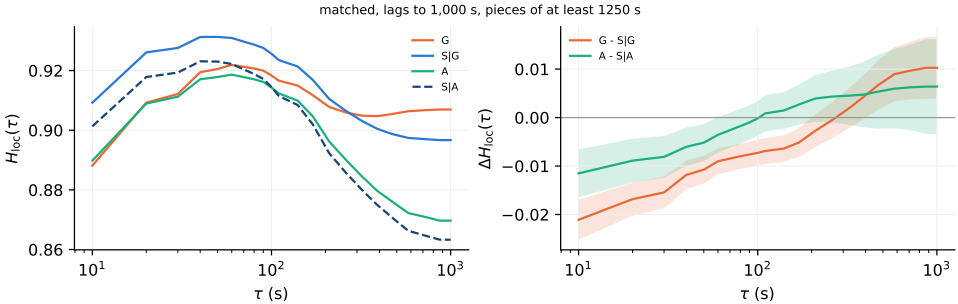
hour of the day ±2 h

same wing class

- Each piece of G is compared with solo segments on its route.
- Controls must match within tolerances, the *calipers*: the first of four sets that keeps 200 pieces per wing level is month ±2, hour ±2, same class: 2 717 pieces of G, 2 474 flights, 1 378 pilots.
- A against its own controls S | A measures who flies in groups.

→ what is a control
→ why the long dataset stops at 2760 s

On the same route the group is still straighter

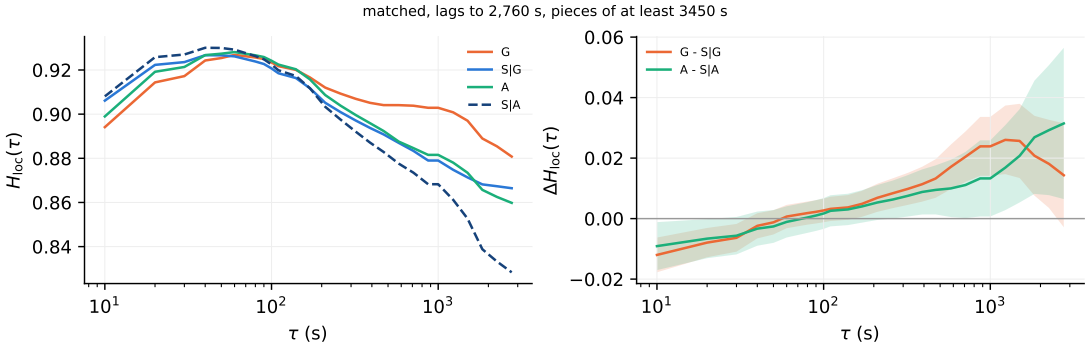


- $\mathrm{G} - \mathrm{S}|\mathrm{G} = +0.010\ [+0.004, +0.017]$ at 1000 s, against controls on the route; the raw $+0.018$ is against all solo flights.
- Selection $\mathrm{A} - \mathrm{S}|\mathrm{A}$: $+0.006\ [-0.003, +0.016]$.

Pieces of at least 1250 s; left: H_{loc} of G, its controls S|G, A and S|A; right: the differences with 90% bands.

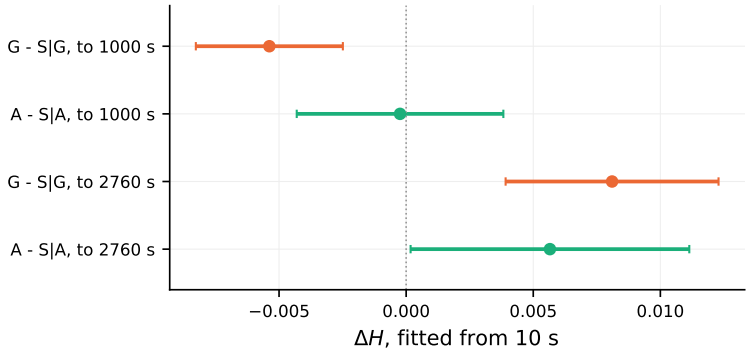
→ the nine quantities
→ robustness

On longer pieces it grows with the lag



- A lag τ needs pieces of at least $\tau/0.8$: for 1840 s, pieces of at least 3450 s.
- $\mathrm{G}-\mathrm{S}|\mathrm{G} = +0.021$ $[+0.008, +0.034]$ at 1840 s; at 1000 s on the same pieces $+0.024$ $[+0.014, +0.034]$.

Fitted H on the same route



$\mathrm{G}-\mathrm{S}|\mathrm{G}$: -0.005 $[-0.008, -0.002]$ to 1000 s and $+0.008$ $[+0.004, +0.012]$ to 2760 s. The sign depends on the lags: H mixes the small deficit at short lags and the gain at long lags.
$\mathrm{A}-\mathrm{S}|\mathrm{A}$: about 0 to 1000 s, $+0.006$ to 2760 s.

On the same route: more climbing; the glide deficit is shared

+3.0 pp
more time in climb
$G - S|G$ [+1.4, +4.5]

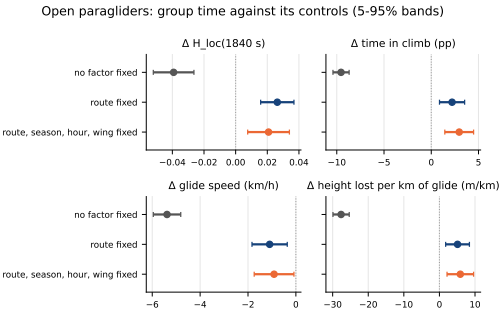
+2.4 pp
net of selection
subtracting $A - S|A$
[+0.8, +4.0]

−0.9
km/h of glide speed,
$G - S|G$
[−1.7, −0.1]

−1.0
km/h for the same pilots alone, $A - S|A$

Only the time in climb survives net of selection (the other differences of the two gaps include 0). The glide deficit is shared by the same pilots alone: it looks like selection, and may explain the opposite sign in question 3. Not tested.

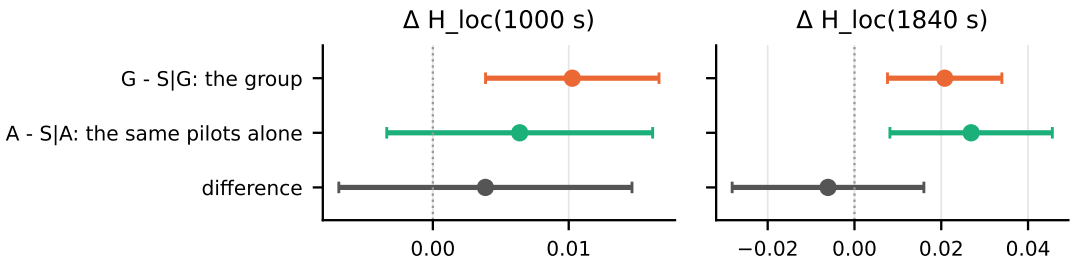
Fixing the route changes the gap; fixing the rest changes little



No factor fixed: controls from any route, solo windows of long flights that climb more than G, so the gap is negative (it is not the raw gap of slide 5). Route fixed: positive. All fixed: about the same. The route carries 98% of what the factors explain at $H_{\mathrm{loc}}(1840\ \mathrm{s})$, 96% for time in climb; season and hour almost nothing.

→ the Shapley split
→ all nine quantities
→ the eight cells

But part of the gap is who flies in groups

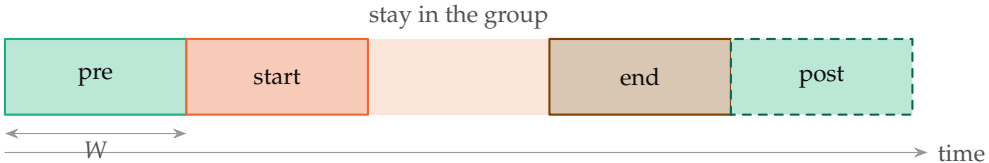


$A - S|A$: the same pilots alone against their controls. At 1840 s they already show the gap: on the same route the group effect is not separable from who flies. We need the same flight: question 3.

Question 3

Does the same flight change when it enters a group, and when it leaves?

The same flight, before, during and after

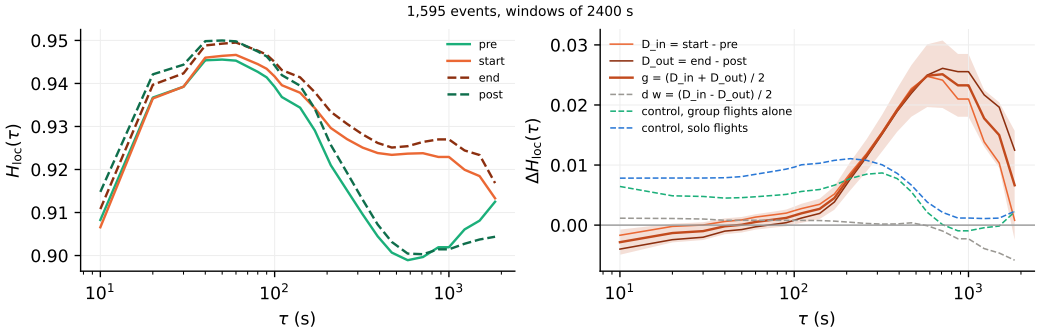


$$D_{\text{in}} = \text{start} - \text{pre}, \quad D_{\text{out}} = \text{end} - \text{post}, \quad g = \tfrac{1}{2}(D_{\text{in}} + D_{\text{out}}), \quad d\,w = \tfrac{1}{2}(D_{\text{in}} - D_{\text{out}})$$

- Same pilot, glider, logger and day: it removes who flies and when.
- g: what the group does; $d\,w$: the drift of the flight over time. An *event*: one group stay with the four windows.
- Controls: the same differences on windows with no group.

Windows of $W = 2400$ s, lags up to 1840 s; open flights.

With windows of 2400 s the flight is straighter in the group

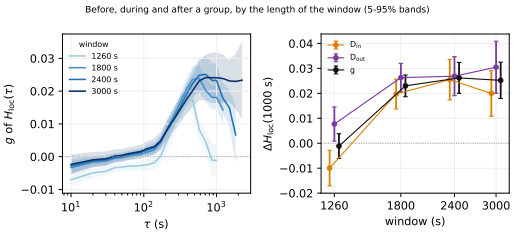


- $g = +0.026$ $[+0.020, +0.032]$ at $H_{\text{loc}}(1000\text{ s})$; $D_{\text{in}} = +0.025$, $D_{\text{out}} = +0.027$.
- Time in climb $+2.8$ $[+2.0, +3.6]$ pp; at 1000 s the control pairs stay near 0.
- $g > 0$ from 50 s, earlier than the raw G − S (170 s).

1 595 events with the four windows.

→ all the differences

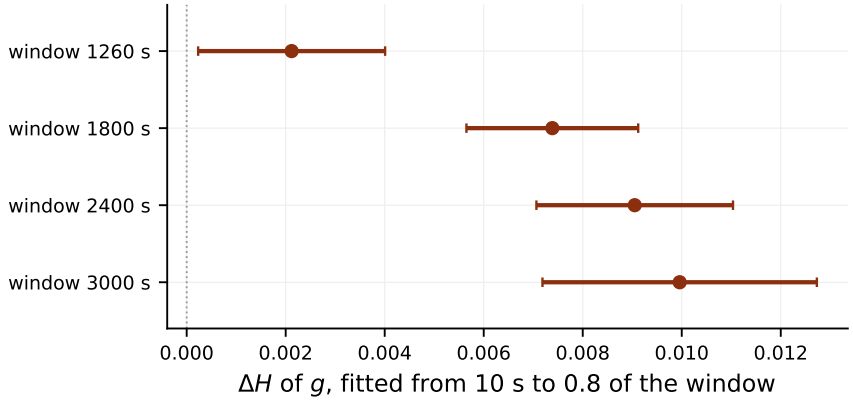
g appears at every window length; $H_{\mathrm{loc}}(1000\ \mathrm{s})$ is a fair quote from 1800 s



- $g(\tau)$ peaks at 390 s for windows of 1260 s ($+0.017$) and at 580–710 s for longer ones ($+0.024$ to $+0.025$): the peak moves with the window, and g falls near its length.
- g at $H_{\mathrm{loc}}(1000\ \mathrm{s})$: -0.001, $+0.023$, $+0.026$, $+0.025$ for 1260, 1800, 2400, 3000 s. At 1260 s the lag is 0.79 of the window: $D_{\mathrm{in}} = -0.010$, $D_{\mathrm{out}} = +0.008$.
- At 1840 s: $g = +0.007$ $[-0.002, +0.016]$ on 2400-s windows, $+0.023$ $[+0.015, +0.031]$ on 3000-s windows; the bands overlap.

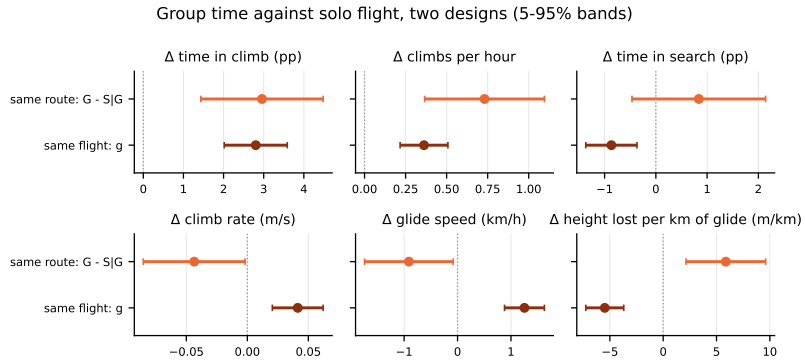
Events with the four windows: 3 549, 2 567, 1 595 and 695.

Fitted H in the same flight



ΔH of g is positive for every window: $+0.002$, $+0.007$, $+0.009$, $+0.010$ for 1260, 1800, 2400, 3000 s (fit from 10 s to 0.8 of the window).

Same route and same flight: some agree, some do not



Inside its group a pilot glides faster and flatter than before and after; on the same route group time glides slower and steeper, but so do the same pilots alone. Net of selection the signs agree with *g*, except for height lost. Not tested.

Take-home

1 In a group the flight is straighter at the long lags

+0.018 $H_{\mathrm{loc}}(1000\ \mathrm{s})$, raw

2 On the same route the group is still straighter; at long lags the pilots alone show it too

+0.010 $H_{\mathrm{loc}}(1000\ \mathrm{s})$, same route

3 The same flight, which removes who flies, is straighter inside its group

+0.026 $H_{\mathrm{loc}}(1000\ \mathrm{s})$, g

+ In all three designs, more time in climb

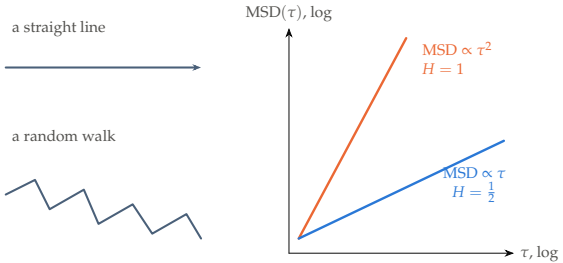
+3 pp +3.3, +3.0, +2.8

Open: glide speed and height lost point the other way in 3 than in 2; the pilots alone show the glide deficit of 2.

Appendix

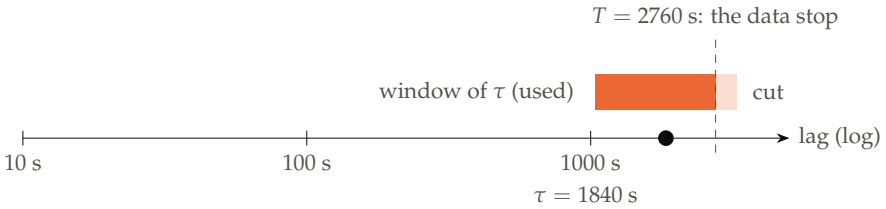
Details, and answers to questions

How we compute H_{loc} in practice



- Per flight, the time-averaged MSD on the 10-s grid of its pieces; the ensemble MSD is the mean over flights, one weight per flight.
- $\text{MSD}(\tau) \sim \tau^{2H}$ on a geometric grid of lags; $H_{\text{loc}}(\tau) = \frac{1}{2} \times$ least-squares slope of $\log \text{MSD}$ against $\log \tau$ over the lags within ± 0.25 decades of τ (from $\tau/1.78$ to $1.78\,\tau$); with fewer than 3 lags, the 3 nearest.
- H: the same fit over all lags, from 10 s to the end of the dataset.

H_{loc} at the edge of the lags: a one-sided estimate



- At the longest lags T of a dataset the window is cut on its upper side: the slope uses only lags below T, so it is centred at a shorter lag than τ and is asymmetric.
- Example: $H_{\mathrm{loc}}(1840\ \mathrm{s})$ on the dataset to $T = 2760$ s uses 1035 to 2760 s instead of 1035 to 3272 s; $H_{\mathrm{loc}}(1000\ \mathrm{s})$ on the dataset to 1000 s uses 562 to 1000 s, but 562 to 1778 s on the dataset to 2760 s: complete.
- All curves of a dataset share T, hence the same cut: differences between populations are like for like, and every bootstrap draw is cut the same way, so the bands include it.

The phase statistics: how each flight counts

Example, glide speed = distance of glides / time in glides, for two flights:

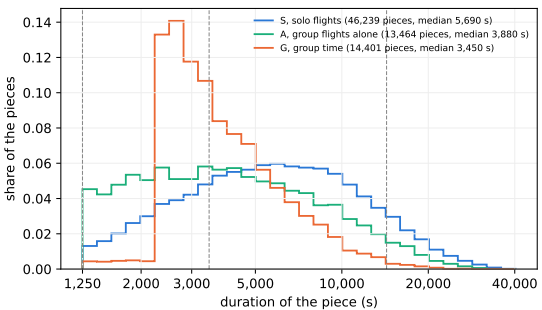
flight	glide distance	glide time	flight time	glide speed
1	20 km	0.5 h	1 h	40 km/h
2	60 km	2 h	4 h	30 km/h

Ratio of sums. $\dfrac{20+60}{0.5+2} = 32$ km/h. Flight 2 weighs four times flight 1: it glides four times longer. The longest flights dominate.

Ours: average over flights, then ratio. $\dfrac{\text{mean}(20/1,\ 60/4)}{\text{mean}(0.5/1,\ 2/4)} = \dfrac{17.5}{0.5} =$ 35 km/h. A flight weighs by its share of time gliding, not by its length.

- With a unit u: $r_u = n_u/T_u$, $q_u = d_u/T_u$, stat $= \sum_u w_u r_u / \sum_u w_u q_u$; n_u, d_u summed over its runs, T_u its flight time.
- The MSD also gives one weight per flight. On the data: time in climb of S 32.9% (ours), 34.2% (ratio of sums).

Datasets and the $\tau/0.8$ rule



A piece of ℓ s gives $(\ell - \tau)/10 + 1$ pairs of origins. Requiring $\ell \geq \tau/0.8$ gives the same share of origins at the longest lag.

- Dataset to T: pieces of at least $T/0.8$ s, at every lag up to T.
- 1000: pieces of 1250 s; 2760: 3450 s; longest (11 450 s): 14 312 s.
- Pieces of G are shorter: a group lasts at least $\tau^* = 38.1$ min.

Why the long dataset stops at 2760 s

caliper (month, hour, wing)	$T = 2250$	2760	3380
±1, ±1, class	311/395	166/231	74/111
±1, ±1, level	327/656	176/372	79/189
±2, ±2, class	513/725	**279/421**	129/228
±2, ±2, level	537/1095	294/650	139/370

pieces of G with a control, beginners/experienced; the rule asks for 200 in each level.

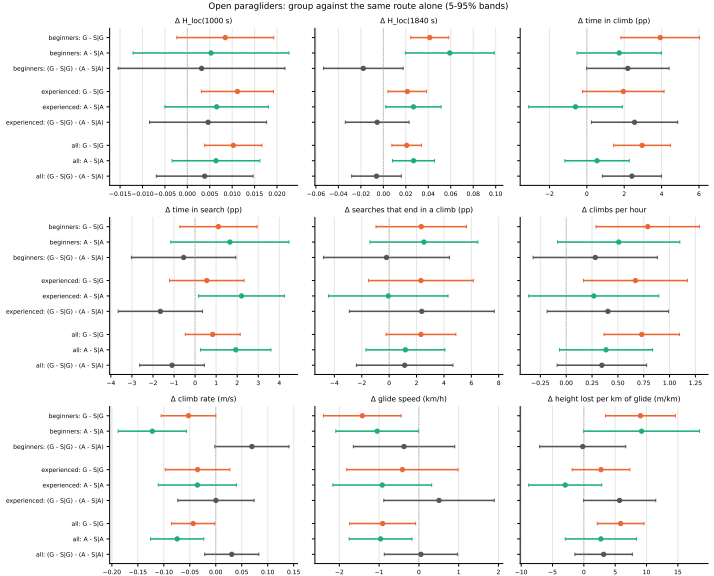
- $H_{\mathrm{loc}}(1840\ \mathrm{s})$ uses the lags within ±0.25 decades: 1035 to 3272 s.
- At $T = 2760$ the window is cut at 2760 s; at 3380 s no caliper reaches 200 pieces per level.

Matching: what is a control

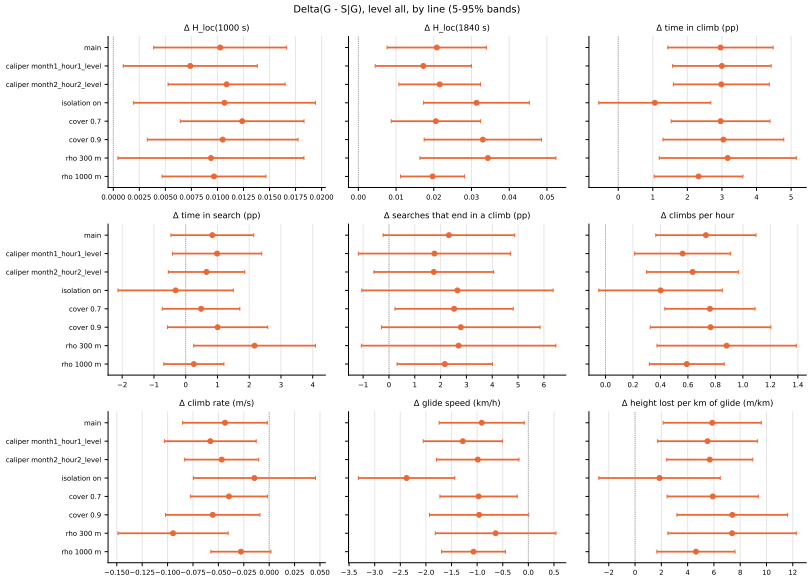
- Candidates: open flights with no contact of any length, on another date.
- Coverage: at least 80% of the points of the piece, one every 30 s, have a pool track point within 500 m, with the same direction.
- Calipers, tried in order: month ± 1 or ± 2, hour ± 1 or ± 2, wing class or level; the chosen one is the first with 200 pieces of G per level.
- Each unit is reweighted to the piece, so a piece with many controls counts once.

Isolation (nobody within 10 km) is off in the main result and is one of the robustness lines.

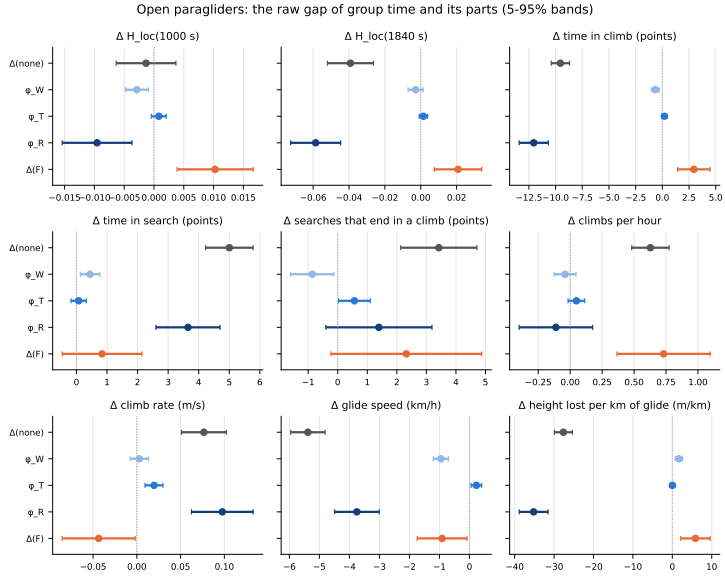
Same route: the nine quantities, by wing level



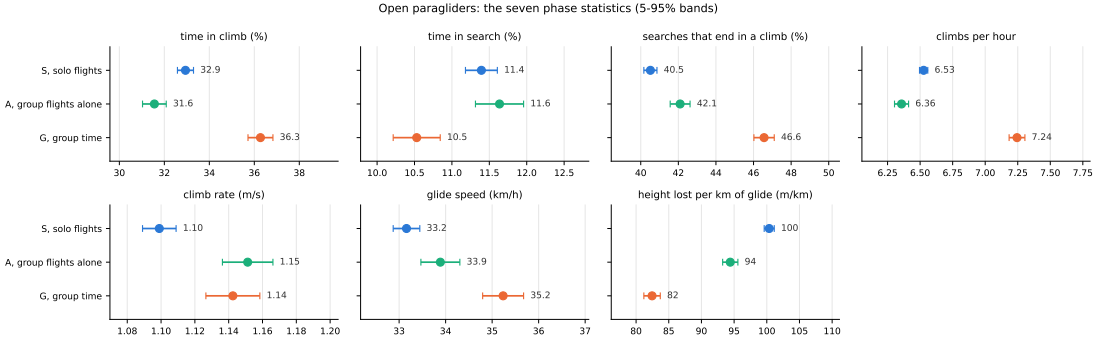
Same route, 8 lines: $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ +0.007 **to** +0.012; $\Delta H_{\mathrm{loc}}(1840\ \mathrm{s})$ +0.017 **to** +0.022 (+0.031 **to** +0.034 **with a shorter long series)**



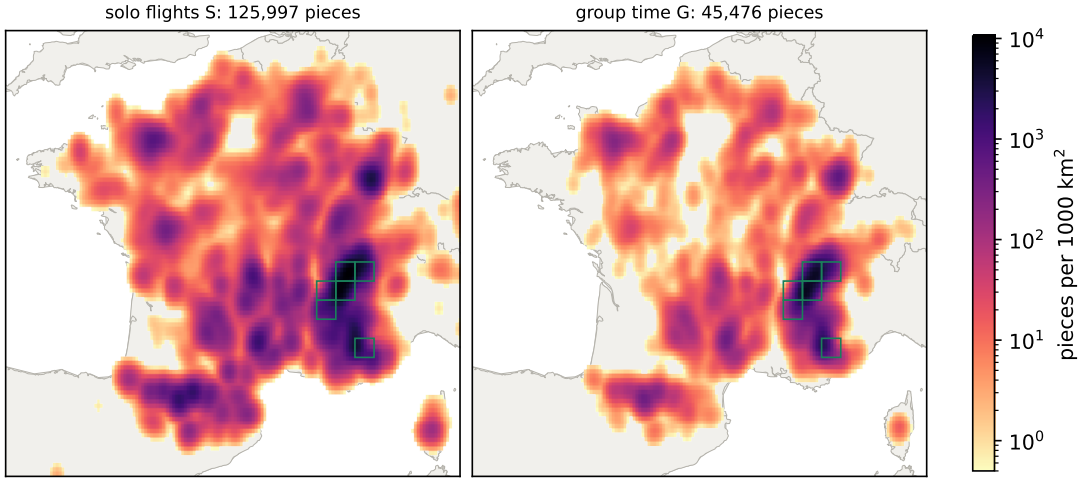
The decomposition: all nine quantities



Raw: the seven phase statistics

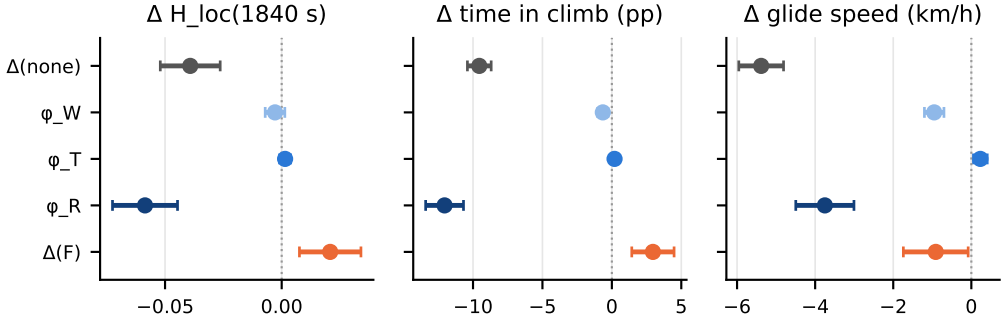


Where solo flights and group time are: pieces per 1000 km^2



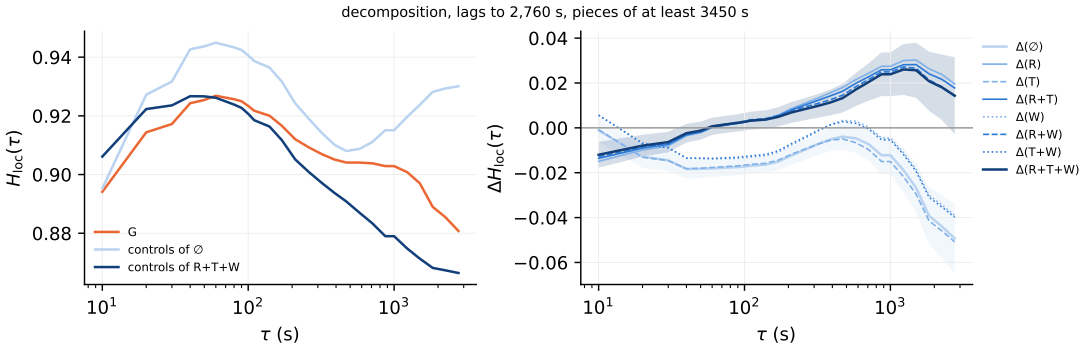
The Shapley split of the raw gap

Open paragliders: the raw gap of group time and its parts (5-95% bands)



Split. route R, season and hour T, wing W: $\Delta(\varnothing) = \Delta(F) + \varphi_R + \varphi_T + \varphi_W$. $\Delta(\varnothing)$ ("none"): no factor fixed; $\Delta(F)$: all fixed. Time in climb: $-9.5 = +3.0 - 12.0 + 0.2 - 0.7$ pp.

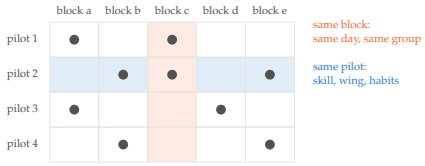
The controls of the decomposition



- Eight cells: each factor fixed or not; $\varphi_R, \varphi_T, \varphi_W$ their Shapley values.
- Controls without the route: windows of solo pieces at least as long as the piece, so from long flights; they climb more and glide faster than S.

Cell ∅: no factor; *F*: route, season and hour, wing.

Where the uncertainty comes from



- The archive is one sample: other seasons would bring other pilots and other days, and every number would change.
- Flights are not independent: those of one pilot share skill, wing and habits; those of one block share the day and the group.
- Pilots and blocks cross: a pilot flies in many blocks, a block holds many pilots.

Block. flights tied by an *episode* of the group identification (groups linked in time through a shared member), or by a *site-day* (takeoffs within 3 km on one day); chained into connected components.

A cell is one pilot in one block, about one flight.

The band: three bootstraps, one formula

- Draw the pilots with replacement; each unit weighs the count of its pilot; recompute the statistic. Its variance over the draws is V_P.
- Same for the blocks (V_B) and for the cells (V_C): 1 000 draws each.

Variance and band. units of one pilot or one block covary, so

$$V = \max\big(V_P + V_B - V_C,\ V_P,\ V_B\big), \qquad \text{band} = \hat{\theta} \pm 1.645\sqrt{V} \quad (90\%).$$

V_C is subtracted once: the pairs of one cell are in both V_P and V_B.

- All the numbers of a section use the same draws: a difference has the band of its own draws.
- Pieces and their controls are drawn together, control flights included: a control that serves many pieces moves them all at once.

Cameron, Gelbach and Miller, J. Bus. Econ. Stat. 29, 238 (2011).

Does the band hold? A check by simulation

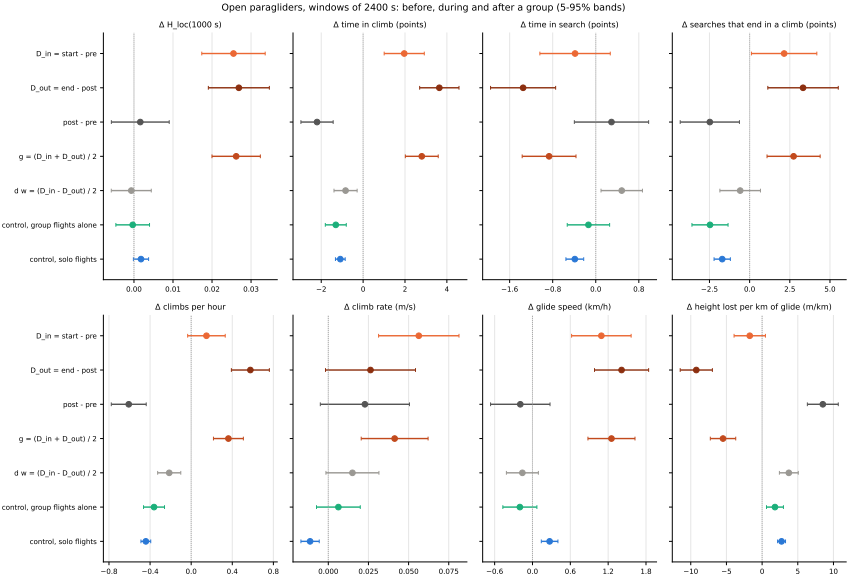
- On the real designs, replace every value by pilot + block + flight + unit effects: no group effect.
- A 90% band must hold 0 in 90% of the replicates.
- The two-way band does so in 88–99%; the bands used before did not (59% when pilots matter).

	raw		same route		same flight	
	new	old	new	old	new	old
unit only	91	90	99	100	90	88
flight only	92	91	89	100	90	90
half flight, half unit	88	85	92	100	88	88
pilot strong	88	59	88	100	89	81
block strong	88	84	88	99	89	88

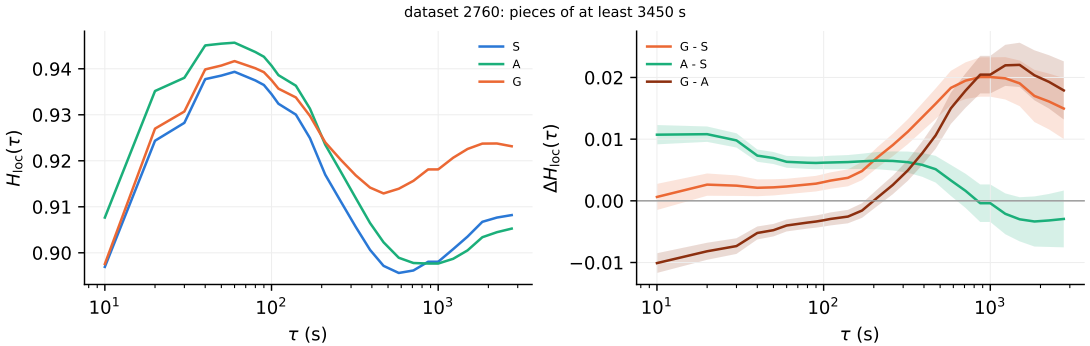
% of replicates whose 90% band holds 0.

Left out of the band: the rules of the matching (varied in the robustness lines) and the dependence on one year (planned).

Before and after, windows of 2400 s: all the differences

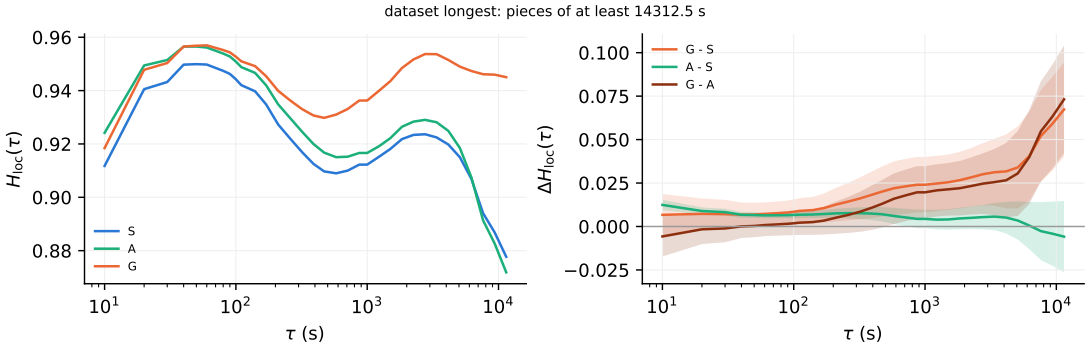


Raw at longer lags: dataset to 2760 s



Pieces of at least 3450 s.

Raw at the longest lags: dataset to 11 450 s



- H_{loc} of S and A falls at the longest lags, that of G does not.
- G keeps few flights there: pieces of at least 14 312 s.